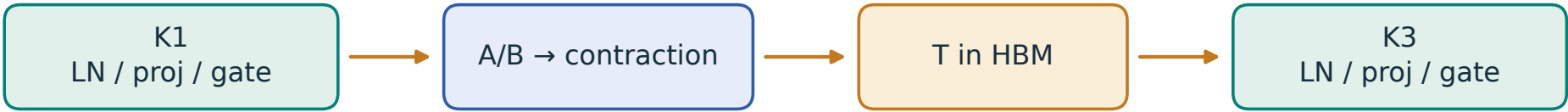


GPU-aware fusion for MiniWorld

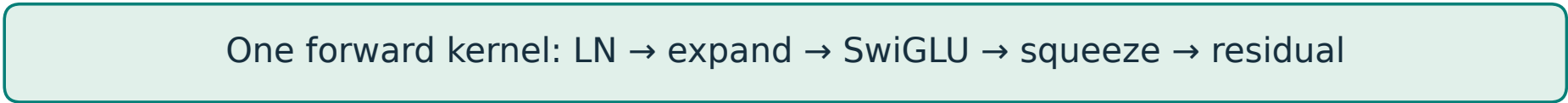
Keep useful intermediates local. Pipeline transfers with compute. Measure forward and backward together.

A / FUSION BOUNDARIES

TriMul · separate global contraction

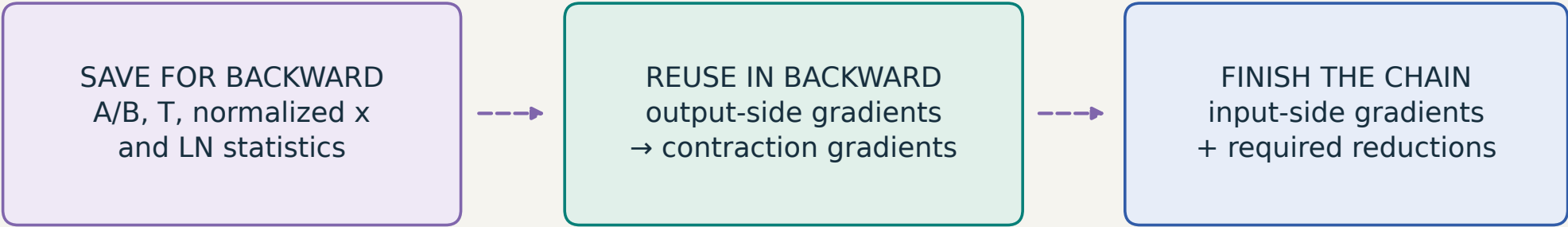
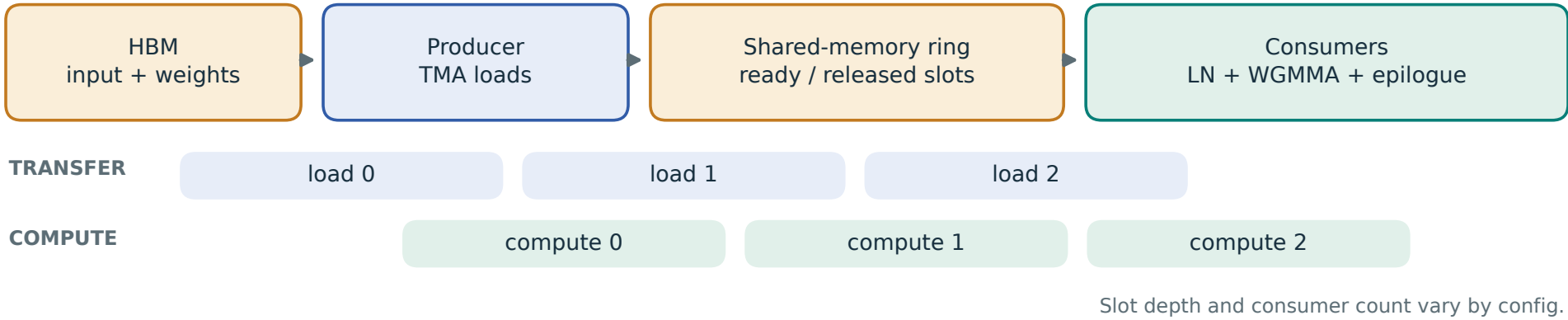


Transition · D128, expansion 4, BF16



B / ALGORITHM AND DATA LIFETIME

INSIDE A K1 / K3 CTA · schematic, not a timing trace



C / MEASURED OPERATOR SPEEDUP

Historical F+B, L384 · family-specific baselines

TriMul **1.61×**

1.634 → 1.018 ms | D128 · T

Transition **1.91×**

1.074 → 0.562 ms | D128, n=4 · X

OPM **2.89×**

5.858 → 2.028 ms | S1024, save O · O

PWA **2.67×**

4.161 → 1.561 ms | S1024 · P

RMSNorm **1.04×**

0.075 → 0.072 ms | D128 · N

LayerNorm **1.12×**

0.276 → 0.246 ms | D384 · N

LNLinear **2.07×**

0.350 → 0.169 ms | D128 → 16; portable path · N

Ratios are not a whole-model speedup.
Portable Norm results do not measure CUDA fusion.

Implementation snapshot: afd54410. Historical checkpoints and baseline identities are documented in the companion figures and evidence bundle.